

Endocrine Physiology:

The Pancreas & the Adrenal Glands

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PHS 220 — Physiology

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§1 ANATOMY OF THE PANCREAS

The pancreas is a lobulated, mixed gland (~15 cm long, 60–100 g) lying retroperitoneally in the upper abdomen, intimately related to the duodenum, portal vein, and superior mesenteric vessels. It is unique in being both an exocrine organ (98% of mass) and an endocrine organ (2% of mass, the islets).

Part	Location	Relations
Head	At L2 near midline; within the C-curve of the duodenum	Posterior: right kidney, IVC, right renal vessels; Medial: CBD and portal vein
Uncinate Process	Extension of head, posterior to portal vein	Posterior to superior mesenteric artery and vein
Neck	Anterior to SMA/SMV and portal vein	Safe surgical plane between neck and portal vessels (no tributaries)
Body	At L1, left of SMA	Anterior to aorta, left adrenal, left kidney; Splenic vessels along superior border
Tail	At T12; reaches splenic hilum	Most mobile part; peritonealised; intraligamentous (in lienorenal ligament)

1.1 Pancreatic Ducts

Duct	Description
Main duct (Wirsung)	Runs the length of the gland; joins the Common Bile Duct (CBD) in 90% of people to form the ampulla of Vater, which opens into the duodenum at the major papilla (guarded by sphincter of Oddi)
Accessory duct (Santorini)	Drains the superior and anterior head; in ~60% enters duodenum 2 cm anterosuperior to the ampulla (at minor papilla)

1.2 Vascular Supply

Region	Arterial Supply	Venous Drainage
Head	Superior pancreaticoduodenal a. (from gastroduodenal a.) + Inferior pancreaticoduodenal a. (from SMA)	Via gastroduodenal trunk → SMV
Body & Tail	Splenic artery branches (transverse pancreatic a., dorsal pancreatic a.)	Into splenic vein → portal vein

§2 EXOCRINE PANCREAS

The exocrine pancreas (acinar cells + centroacinar cells) secretes 1.5–2 L/day of a bicarbonate-rich, enzyme-rich juice essential for digestion. This juice is clear, watery, and alkaline (pH 8.0–8.3), neutralising the acidic chyme entering the duodenum from the stomach.

Component	Source	Function
Bicarbonate (HCO_3^-)	Centroacinar cells (stimulated by secretin)	Neutralises gastric acid; provides optimal pH for pancreatic enzymes (~7-8)
Amylase	Acinar cells (active form secreted)	Digests starch → maltose, oligosaccharides
Lipases (pancreatic lipase, colipase, phospholipase A2)	Acinar cells	Digest triglycerides → fatty acids + monoglycerides; colipase required for lipase activity in presence of bile
Proteases (trypsinogen, chymotrypsinogen, proelastase, procarboxypeptidase)	Acinar cells (as inactive zymogens)	Activated in duodenum; trypsin is master activator (enterokinase activates trypsinogen); digest proteins
Ribonucleases & DNases	Acinar cells	Digest nucleic acids from food

2.1 Phases of Secretion

Phase	Trigger	Mechanism	Output
Cephalic (~20%)	Thought, sight, smell, taste, anticipation of food	Vagal cholinergic stimulation; VIP release	Enzyme-rich juice; some HCO_3^-
Gastric (~10%)	Gastric distension; exposure to nutrients	Vagovagal reflex; gastrin release	Mainly enzyme-rich secretion
Intestinal (~70%)	Acid chyme entering duodenum; fat and protein	CCK (from I-cells) → enzyme secretion; Secretin (from S-cells) → HCO_3^- secretion. Also inhibited by pancreatic polypeptide and peptide YY	Largest phase; enzyme and HCO_3^- rich

§3 ENDOCRINE PANCREAS (ISLETS OF LANGERHANS)

The Islets of Langerhans are ~1-2% of total pancreatic mass but receive ~20% of the pancreatic blood flow — reflecting their high metabolic activity and the need for hormones to enter the portal circulation. There are approximately 1 million islets in the adult pancreas.

Cell Type	Proportion	Distribution	Hormone Produced	Function
β (Beta) cells	60-80%	Core of islet; evenly distributed throughout gland	Insulin	Lower blood glucose; anabolic — promotes storage
α (Alpha) cells	15-20%	Mantle of islet; predominant in body and tail	Glucagon	Raise blood glucose; catabolic — promotes mobilisation

Cell Type	Proportion	Distribution	Hormone Produced	Function
δ (Delta) cells	5-10%	Scattered; paracrine position adjacent to α and β	Somatostatin	Inhibit both insulin and glucagon; paracrine regulation
F cells (PP cells)	15-20%	Predominantly in head of pancreas	Pancreatic Polypeptide (PP)	Inhibit exocrine secretion; modulate GI motility
ε (Epsilon) cells	<1%	Rare	Ghrelin	Stimulate hunger; modulate insulin secretion

§4 INSULIN: STRUCTURE, SECRETION & ACTION

4.1 Chemistry of Insulin

Insulin is a 51-amino acid polypeptide consisting of two chains — the A chain (21 aa) and B chain (30 aa) — linked by two disulphide bonds (and one intrachain disulphide in the A chain). It is derived from a single-chain precursor:

- Preproinsulin (110 aa) is synthesised on ribosomes → signal peptide cleaved in ER → Proinsulin
- Proinsulin (86 aa) is folded and disulphide bonds form in the ER → transported to Golgi
- In secretory granules, C-peptide (31 aa connecting peptide) is cleaved by prohormone convertases → equimolar insulin + C-peptide secreted
- C-peptide measurement is used clinically to assess endogenous insulin secretion (not suppressed by exogenous insulin treatment)

4.2 Regulation of Insulin Release

Stimulus	Effect on Insulin	Mechanism
↑ Blood glucose (primary)	Potent stimulator	Glucose enters β-cell via GLUT-2 → phosphorylated → ↑ ATP/ADP ratio → closes K-ATP channels → membrane depolarisation → Ca ²⁺ influx via L-type channels → exocytosis of insulin granules
Amino acids (arginine, leucine)	Stimulator	Especially post-protein meal; synergises with glucose; prevents hypoglycaemia from amino acid-stimulated glucagon
GIP and GLP-1 (incretins)	Stimulator (glucose-dependent)	Released from gut in response to meal; amplify glucose-stimulated insulin secretion — basis of GLP-1 receptor agonist therapy (e.g., semaglutide)
CCK	Mild stimulator	Acts on β-cell CCK receptors
Parasympathetic NS (ACh)	Stimulator	Muscarinic M3 receptors → IP3/Ca ²⁺ → insulin release
Glucagon	Stimulator	Via cAMP in β-cells (paracrine)

Stimulus	Effect on Insulin	Mechanism
Ketoacids	Stimulator	Prevents excess lipolysis during ketosis
Sympathetic NS (β_2 -AR)	Stimulator	Mild; less important than α_2 inhibition
Sympathetic NS (α_2 -AR)	Potent inhibitor	During stress → maintains blood glucose for brain and muscles; opens K-ATP channels
Somatostatin	Inhibitor	Paracrine from δ -cells; reduces insulin and glucagon (meal termination)
Hyperglycaemia (sustained)	Biphasic response	First phase: rapid release of preformed granules (2-5 min); Second phase: new synthesis and release

4.3 Insulin Receptor & Signalling

The insulin receptor is a heterotetrameric transmembrane glycoprotein ($\alpha_2\beta_2$) with intrinsic tyrosine kinase activity. The α subunits are extracellular and contain the insulin-binding domain; the β subunits span the membrane and contain the kinase domain.

- Insulin binds α subunits → conformational change → trans-autophosphorylation of β subunit tyrosines
- Activated receptor phosphorylates Insulin Receptor Substrate proteins (IRS-1, IRS-2)
- IRS → PI3K → PIP3 → PDK1 → Akt/PKB → multiple downstream effects (GLUT4 translocation, glycogen synthesis, protein synthesis, anti-apoptosis)
- IRS → Ras → Raf → MEK → MAPK (ERK) → gene expression, cell growth, proliferation

4.4 Metabolic Actions of Insulin — Fed State

Target Tissue	Anabolic Action of Insulin
Liver	↑ Glycogen synthesis (activates glycogen synthase); ↑ Glycolysis; ↑ Lipogenesis (↑ fatty acid synthesis from glucose); ↓ Gluconeogenesis; ↓ Glycogenolysis; ↓ Ketogenesis
Skeletal Muscle	↑ GLUT4 translocation → ↑ glucose uptake; ↑ Glycogen synthesis; ↑ Protein synthesis; ↓ Protein catabolism
Adipose Tissue	↑ GLUT4 → ↑ glucose uptake; ↑ Lipogenesis (lipoprotein lipase activation → ↑ FFA uptake); ↓ Lipolysis (inhibits hormone-sensitive lipase)
All cells	↑ K^+ uptake (via Na/K-ATPase) → clinical use: dextrose+insulin for hyperkalaemia
Brain	Insulin-INDEPENDENT glucose uptake (GLUT1, GLUT3); some insulin sensitivity in hypothalamus (satiety signalling)

§5 GLUCAGON: STRUCTURE, SECRETION & ACTION

Glucagon is a 29-amino acid single-chain polypeptide secreted by α -cells. It is the primary counter-regulatory hormone to insulin, acting to raise blood glucose during fasting and stress. Its

secretion is the mirror image of insulin secretion.

Stimulus	Effect on Glucagon
↓ Blood glucose (hypoglycaemia)	Potent stimulator — primary physiological trigger
Amino acids (especially arginine)	Stimulator — prevents hypoglycaemia after high-protein meal
Sympathetic NS (β_2 -AR on α -cells)	Stimulator — stress response to ensure glucose availability
Exercise	Stimulator — ↑ glucose requirements of muscle
↑ Blood glucose	Inhibitor
Insulin	Inhibitor — paracrine from adjacent β -cells
Somatostatin	Inhibitor — paracrine from δ -cells
Free fatty acids	Inhibitor

5.1 Mechanism & Actions of Glucagon — Fasting State

Glucagon binds its GPCR (Gs-coupled) on hepatocytes → ↑ cAMP → ↑ PKA activity. It also activates the Gq/IP3 pathway in the liver. Main target: **liver** (where it exerts its dominant metabolic effects).

Target Tissue	Glucagon Action	Metabolic Consequence
Liver (primary)	↑ Glycogenolysis (↑ phosphorylase kinase → glycogen → glucose-1-P)	Rapid glucose release into blood
Liver	↑ Gluconeogenesis (↑ PEPCK expression)	Sustained glucose production from amino acids, lactate, glycerol
Liver	↓ Glycolysis (inhibits PFK-2 → ↓ fructose-2,6-bisphosphate)	Prevents glucose consumption by liver during fasting
Liver	↑ Ketogenesis (↑ β -oxidation, ↑ mitochondrial entry via ↓ malonyl-CoA)	Ketone bodies as alternative fuel (brain during prolonged fast)
Adipose	↑ Lipolysis (activates hormone-sensitive lipase via PKA → ↑ FFAs)	Provides substrate for hepatic gluconeogenesis and ketogenesis
Muscle (minor)	↑ Protein catabolism → amino acids	Substrate for gluconeogenesis

§6 COUNTER-REGULATORY HORMONES & GLUCOSE HOMEOSTASIS

Several hormones work in concert to oppose the hypoglycaemic effect of insulin and maintain blood glucose during fasting and stress. Understanding these is critical for managing diabetic hypoglycaemia.

Hormone	Source	Anti-insulin Mechanism	Relative Importance
Glucagon	Pancreatic α -cells	$\uparrow$ Hepatic glycogenolysis and gluconeogenesis; $\uparrow$ lipolysis	★★★★★ Primary counter-regulatory hormone; first-line defence
Adrenaline (Epinephrine)	Adrenal medulla	$\uparrow$ Glycogenolysis (liver & muscle); $\uparrow$ lipolysis; inhibits insulin secretion via α_2 -AR	★★★★ Critical especially when glucagon deficient; basis of glucagon injection for severe hypo
Cortisol	Adrenal cortex	$\uparrow$ Gluconeogenesis; $\uparrow$ protein catabolism; $\uparrow$ lipolysis; permissive for other counter-regulatory hormones; reduces insulin sensitivity	★★★ Sustained hypoglycaemia; essential for recovery over hours
Growth Hormone	Anterior pituitary	$\uparrow$ Lipolysis; $\uparrow$ gluconeogenesis; reduces glucose uptake; anti-insulin at peripheral tissues	★★★ Acts over hours; impaired recovery in GH deficiency
Noradrenaline	Sympathetic nerves + adrenal medulla	Similar to adrenaline; hepatic glycogenolysis; inhibits insulin	★★ Supports adrenaline response

§7 ANATOMY OF THE ADRENAL GLANDS

The adrenal (suprarenal) glands are paired retroperitoneal endocrine organs, each weighing approximately 4–5 g, situated on the superior poles of the kidneys within Gerota's fascia. They are essential for life, producing hormones that regulate metabolism, electrolyte balance, stress response, and sexual characteristics.

Feature	Right Adrenal	Left Adrenal
Shape	Pyramidal (triangular)	Crescentic (semilunar)
Position	More medial; closer to IVC	More lateral; higher in position
Venous drainage	Short right adrenal vein $\rightarrow$ IVC directly (surgical risk)	Left adrenal vein $\rightarrow$ left renal vein $\rightarrow$ IVC
Arterial supply	Three sources each: Superior (inf. phrenic a.), Middle (aorta), Inferior (renal a.)	Same three-source pattern

7.1 Adrenal Zones — Cortex & Medulla

Zone	% of Cortex	Hormone	Mnemonic	Enzyme Unique To Zone
Zona Glomerulosa (outer)	15%	Mineralocorticoids (Aldosterone 90%)	GFR: Glomerulosa = Goodies (salt)	Aldosterone synthase (CYP11B2)
Zona Fasciculata (middle)	75%	Glucocorticoids (Cortisol 95%)	GFR: Fasciculata = Fat (sugar)	17 α -hydroxylase; 11 β -hydroxylase (CYP11B1)
Zona Reticularis (inner)	10%	Androgens (DHEA, androstenedione)	GFR: Reticularis = Remains (sex)	17,20-lyase
Medulla (innermost)	N/A (chromaffin cells)	Catecholamines (Adrenaline 80%, Noradrenaline 20%)	Modified sympathetic neurons	PNMT (N-methyltransferase: NA $\rightarrow$ Adrenaline, induced by cortisol)

§8 GLUCOCORTICOIDS (CORTISOL)

Cortisol accounts for approximately 95% of glucocorticoid activity in humans. It is synthesised in the zona fasciculata from cholesterol via a series of cytochrome P450-catalysed steps. Understanding cortisol physiology is essential because glucocorticoids are among the most widely prescribed drugs in medicine.

8.1 Regulation — HPA Axis

- Hypothalamus secretes **CRH** (Corticotropin-Releasing Hormone) into portal blood in a pulsatile, circadian pattern (peak: early morning, 6–8 AM).
- CRH stimulates anterior pituitary corticotrophs to secrete **ACTH** (Adrenocorticotrophic Hormone, derived from POMC).
- ACTH binds MC2R on zona fasciculata cells $\rightarrow$ $\uparrow$ cAMP $\rightarrow$ stimulates all steps of cortisol synthesis (acute: cholesterol mobilisation; chronic: gene expression of steroidogenic enzymes).
- Rising **cortisol** provides negative feedback to both hypothalamus ($\downarrow$ CRH) and anterior pituitary ($\downarrow$ ACTH).
- Stress inputs (hypoglycaemia, pain, sepsis, surgery, burns) override the negative feedback, driving a powerful stress cortisol response.

8.2 Actions of Cortisol

System	Cortisol Effect
Metabolism — Carbohydrate	$\uparrow$ Gluconeogenesis ($\uparrow$ PEPCK); $\uparrow$ glycogen deposition in liver; $\downarrow$ peripheral glucose uptake $\rightarrow$ hyperglycaemia ("steroid diabetes")
Metabolism — Protein	$\uparrow$ Protein catabolism in muscle, skin, bone $\rightarrow$ amino acids for gluconeogenesis; in liver $\uparrow$ protein synthesis
Metabolism — Fat	$\uparrow$ Lipolysis ($\rightarrow$ FFAs for gluconeogenesis); excess cortisol $\rightarrow$ redistribution (truncal obesity, buffalo hump, moon face, thin limbs)

System	Cortisol Effect
Anti-inflammatory	Stabilises lysosomal membranes; ↓ capillary permeability; inhibits phospholipase A2 (via lipocortin); ↓ arachidonic acid → ↓ prostaglandins, leukotrienes; ↓ cytokine production
Immunosuppression	↓ Lymphocytes (lymphopaenia); ↓ eosinophils; ↓ macrophage activity; ↑ neutrophils (demargination); ↓ IgG (chronic use)
Fluid & Electrolytes	Some mineralocorticoid activity → Na ⁺ retention, K ⁺ loss, ↑ free water clearance; ↑ uric acid
Haematological	↑ RBC (↑ EPO); ↑ Neutrophils; ↓ Eosinophils; ↓ Lymphocytes (WENE mnemonic for excess cortisol)
CNS	Maintains mood; excess → psychosis, depression; deficiency → fatigue, depression
Skeletal	Excess → osteoporosis (↓ osteoblasts, ↑ osteoclasts, ↓ Ca ²⁺ absorption, ↓ sex hormones)
Permissive actions	Required for normal vascular tone and catecholamine sensitivity; essential for renal free water clearance

8.3 Disorders of Glucocorticoids

Disorder	Definition	Cause	Key Features	Investigations
Cushing's Syndrome	Excess cortisol (any cause)	Exogenous steroids (most common); Cushing's disease (pituitary ACTH adenoma 70%); Ectopic ACTH (15%); Adrenal adenoma/carcinoma (15%)	Central obesity, moon face, buffalo hump, purple striae, proximal myopathy, osteoporosis, hypertension, hyperglycaemia, hirsutism, thin skin, easy bruising, acne, depression	Screening: 24h urinary cortisol; overnight dexamethasone suppression test. Localisation: ACTH level + MRI pituitary + CRH test + inferior petrosal sinus sampling
Addison's Disease (Primary AI)	Adrenocortical insufficiency — all 3 zones affected	Autoimmune (70%); TB; metastases; infarction; surgical	Fatigue, weight loss, anorexia, N/V, postural hypotension, hyperpigmentation (↑ ACTH/MSH), hyponatraemia, hyperkalaemia, hypoglycaemia	Low morning cortisol; High ACTH; Failed short Synacthen test (< 550 nmol/L at 30 min); positive adrenal antibodies
Congenital Adrenal Hyperplasia (CAH)	21-hydroxylase deficiency (90%)	Autosomal recessive enzyme defect	Cortisol deficiency + ACTH excess → adrenal hyperplasia; mineralocorticoid deficiency + androgen excess; female virilisation; salt-wasting crisis in neonates	Very high 17-OH-progesterone; low cortisol; low/absent aldosterone

§9 MINERALOCORTICOIDS (ALDOSTERONE)

Aldosterone accounts for ~90% of mineralocorticoid activity and is synthesised exclusively in the zona glomerulosa (which lacks 17 α -hydroxylase). It is the principal regulator of sodium balance, extracellular fluid volume, and blood pressure through its actions on the distal nephron.

9.1 Regulation of Aldosterone Secretion

Stimulus	Mechanism	Potency
RAAS — Angiotensin II	$\downarrow$ BP $\rightarrow$ Renin (JGA) $\rightarrow$ Angiotensinogen $\rightarrow$ Ang I $\rightarrow$ Ang II (ACE) $\rightarrow$ Aldosterone synthesis	Most important (primary regulator)
$\uparrow$ Serum K ⁺	Direct depolarisation of glomerulosa cells $\rightarrow$ Ca ²⁺ influx $\rightarrow$ aldosterone synthesis	Very potent (small $\uparrow$ in K ⁺ $\rightarrow$ large $\uparrow$ in aldosterone)
ACTH	Via cAMP; less important chronically than RAAS or K ⁺	Permissive (needed for baseline, not primary regulator)
$\downarrow$ Na ⁺ (hyponatraemia)	Via macula densa $\rightarrow$ $\uparrow$ renin $\rightarrow$ RAAS	Indirect via RAAS
$\uparrow$ Na ⁺ or $\downarrow$ K ⁺	Negative feedback — $\downarrow$ renin $\rightarrow$ $\downarrow$ Ang II $\rightarrow$ $\downarrow$ aldosterone	Inhibitory

9.2 Actions of Aldosterone

Aldosterone acts on the **distal convoluted tubule and collecting duct** of the kidney (principal cells). As a steroid hormone, it binds intracellular mineralocorticoid receptors (MR) $\rightarrow$ translocate to nucleus $\rightarrow$ bind aldosterone response elements $\rightarrow$ transcription of sodium transport proteins.

- $\uparrow$ Expression of ENaC (epithelial Na⁺ channels) on apical membrane $\rightarrow$ $\uparrow$ Na⁺ reabsorption into cell
- $\uparrow$ Na⁺/K⁺-ATPase on basolateral membrane $\rightarrow$ actively moves Na⁺ into blood, K⁺ into cell
- $\uparrow$ ROMK channels on apical membrane $\rightarrow$ K⁺ secreted into tubular lumen $\rightarrow$ urinary K⁺ excretion
- Na⁺ reabsorption creates electrochemical gradient $\rightarrow$ H⁺ also secreted into tubule $\rightarrow$ urine acidified, blood alkalinised
- Water follows Na⁺ osmotically $\rightarrow$ $\uparrow$ ECF volume (but aldosterone does NOT directly regulate water channels)

ALDOSTERONE DISORDERS

Primary Hyperaldosteronism (Conn's Syndrome): Adrenal adenoma or bilateral hyperplasia $\rightarrow$ autonomous aldosterone $\rightarrow$ hypertension, hypokalaemia, metabolic alkalosis, suppressed renin and Ang II. Accounts for ~10% of hypertension. Diagnosed by aldosterone:renin ratio.

Without aldosterone: Sodium and water loss $\rightarrow$ dehydration, hypotension (Addison's crisis).

Excess aldosterone $\rightarrow$ hypokalaemia: Muscle weakness; cardiac arrhythmias (U-waves on ECG).

Deficit aldosterone $\rightarrow$ hyperkalaemia: Dangerous cardiac toxicity (peaked T-waves, VF).

§10 ADRENAL ANDROGENS & CATECHOLAMINES

10.1 Adrenal Androgens

The zona reticularis produces weak androgens — DHEA (dehydroepiandrosterone), DHEA-sulphate (DHEAS), and androstenedione. These are converted peripherally to the more potent testosterone (and oestradiol in females). In women, the adrenal provides the majority of circulating androgens.

Androgen	Relative Potency	Clinical Relevance
DHEA/DHEAS	Weak (~5% of testosterone)	Excess in CAH → female virilisation; also in adrenal carcinoma
Androstenedione	Weak (~10% of testosterone)	Converted to testosterone in peripheral tissues and oestrone in fat
Testosterone (trace)	Potent	Adrenal contribution important in women (libido, muscle); also converted to DHT and oestradiol

10.2 Adrenal Medullary Catecholamines

The adrenal medulla is a modified sympathetic ganglion containing chromaffin cells that synthesise and store catecholamines (adrenaline 80%, noradrenaline 20%, trace dopamine). Unlike neurons, chromaffin cells release hormones into the blood rather than across a synapse, producing systemic effects.

Step	Substrate → Product	Enzyme	Site
1	Tyrosine → L-DOPA	Tyrosine hydroxylase (rate-limiting)	Cytoplasm
2	L-DOPA → Dopamine	DOPA decarboxylase	Cytoplasm
3	Dopamine → Noradrenaline	Dopamine β-hydroxylase	Secretory granules
4	Noradrenaline → Adrenaline	PNMT (Phenylethanolamine-N-methyl transferase)	Cytoplasm (cortisol from adrenal cortex induces PNMT)

Receptor	Location	Effect of Adrenaline / Noradrenaline
α1	Vascular smooth muscle; pupil dilator; liver	Vasoconstriction; mydriasis; glycogenolysis
α2	Presynaptic nerve endings; pancreatic β-cells	↓ NE release; ↓ insulin secretion
β1	Heart; kidney JGA	↑ HR, ↑ contractility (chronotropy, inotropy); ↑ renin
β2	Bronchi; skeletal muscle vasculature; β-cells; liver	Bronchodilation; vasodilation; ↑ glycogenolysis/gluconeogenesis; ↑ insulin
β3	Adipose tissue	Lipolysis; thermogenesis

PHAEOCHROMOCYTOMA

A catecholamine-secreting tumour of chromaffin cells (adrenal medulla 90%; extra-adrenal = paraganglioma). Classical presentation: episodic hypertension, headache, sweating, palpitations (the '3 Ps'). Sustained hypertension in 50%.

Rule of 10s: 10% malignant, 10% bilateral, 10% extra-adrenal, 10% familial (MEN2, VHL, NF1).

Diagnosis: 24h urinary catecholamines/metanephrines; plasma metanephrines (most sensitive).

Treatment: α -blockade FIRST (phenoxybenzamine), THEN β -blockade (to avoid hypertensive crisis from unopposed α); then surgical resection.